

Visual Representation of Crime News

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Most media research focuses on text

What the literature usually analyses

Iran War Live Updates: Tehran and U.S. Offer Conflicting Messages on State of Negotiations

 Summarize

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Updated May 6, 2026, 6:38 p.m. ET11 minutes ago

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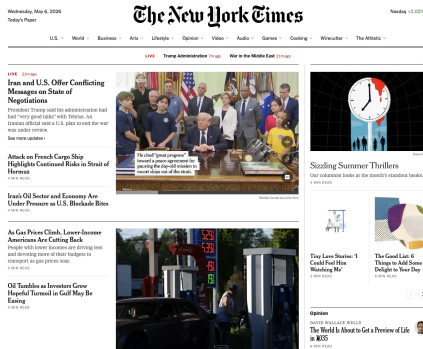
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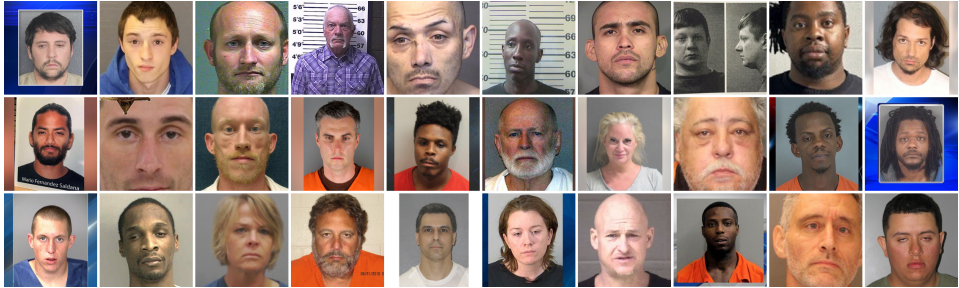
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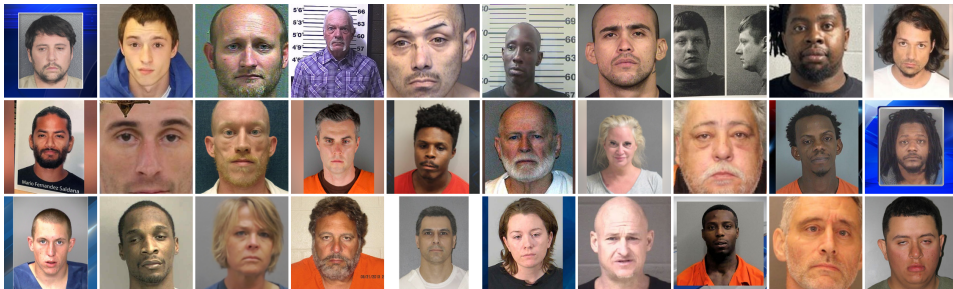
What a media readers see



This paper: a particular visual element of crime news

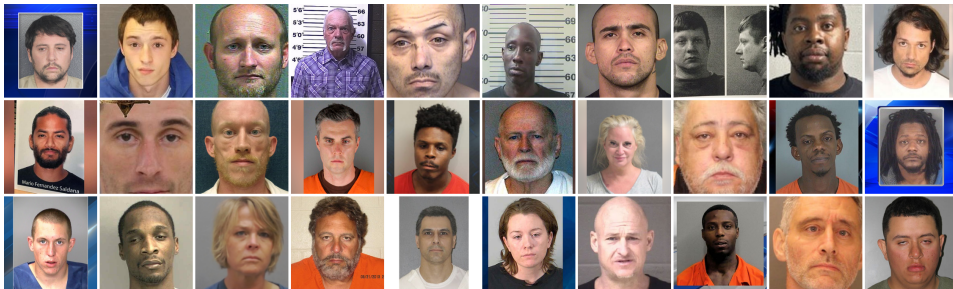


This paper: a particular visual element of crime news



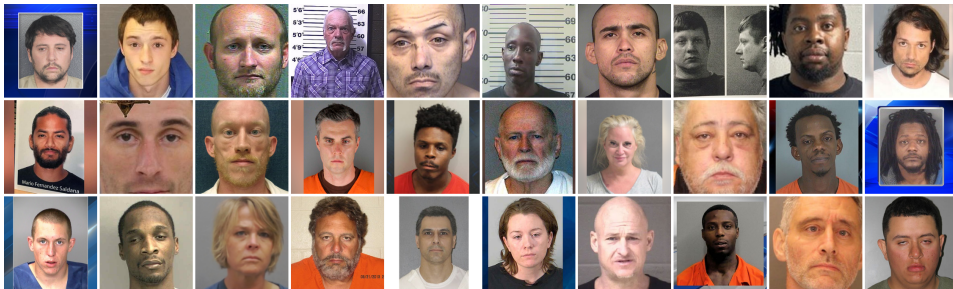
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This paper: a particular visual element of crime news



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- Shape the visual impression of who commits crime.

This paper: a particular visual element of crime news



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- Shape the visual impression of who commits crime.

How are mugshots used in US news media?
Are there racial disparities in mugshot publication?

Why crime news

Contribution

Media Slant and Persuasive Effects of News and Images

1. News content is often slanted following ideological or market-driven incentives
Gentzkow and Shapiro (2008), Gentzkow and Shapiro (2010), Durante and Knight (2012), and Djourelova et al. (2025)
2. Media content affects attitudes and behavior
DellaVigna and Kaplan (2007), Ladd and Lenz (2009), Chiang and Knight (2011), Puglisi and Snyder (2015), Martin and Yurukoglu (2017), Grossman et al. (2022), Djourelova (2023), and Marangon (2026)
3. Few evidence on the persuasive effects of images
Peng (2018), Ash et al. (2021), Macchi (2023), Torres (2024), Caprini (2024), and Ciccarone (2025)

Contribution

Media Slant and Persuasive Effects of News and Images

1. News content is often slanted following ideological or market-driven incentives
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3. Few evidence on the persuasive effects of images

Crime, Media, and Images

1. Extensive evidence of racial disparities in criminal justice
Knowles et al. (2001), Abrams et al. (2012), Anwar et al. (2012), Rehavi and Starr (2014), and Arnold et al. (2018)
2. Higher crime coverage increases perceptions of crime and criminal justice outcomes
Lim et al. (2015), Mastroiocco and Minale (2018), Ash and Poyker (2024), Mastroiocco and Ornaghi (2025), and Moreno-Medina et al. (2025)
3. Visual cues are important determinants of criminal justice outcomes
Ludwig and Mullainathan (2024)

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2. Higher crime coverage increases perceptions of crime and criminal justice outcomes
3. Visual cues are important determinants of criminal justice outcomes

→ Measure visual slant in crime news and investigate racial disparities in mugshot publication by news media

Outline

1. Motivation

2. Methods

3. Initial findings

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Data: crime articles and their images

- **Media articles** about crime in the U.S. from NewsAPI.ai in 2015–2024
 - Contains information about FILL IN and image URLs
- **Articles about crime events** isolated using an ML classifier (XGBoost + TF-IDF) based on an LLM-labeled and manually validated sample.
- **Images** scraped where available.

Image classification

Mugshot detection

- VLM labels a sample of crime-article images as mugshot vs. non-mugshot.
- Sample manually validated.
- Fine-tune an EfficientNet model on those labels and apply it corpus-wide.

details

Image classification

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Race classification

- Identify faces using face recognition (MTCNN FaceNet).
- Train a per-face race classifier using the FairFace dataset (MLP with face embeddings as inputs).
- Apply it to mugshot images.

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Face deduplication

- Extract face embeddings from mugshots
- Create nearest neighbor graph
- Chinese-Whispers clustering at cosine $t = 0.90$

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Text classification

Stage of the case

- LLM labels stage on sample
- Distilled to TF-IDF + multinomial logistic regression
- Applied to the full crime corpus

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Crime type

- Same LLM-label-then-distill recipe, multi-label setting
- One-vs-rest logistic regression on TF-IDF features
- Reported as top-1 by default; multi-label sensitivity available

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Outlet

- LLM classifies outlets as news / non-news and local / national.
- Outlet headquarters extracted with LLM + web-search tool

details

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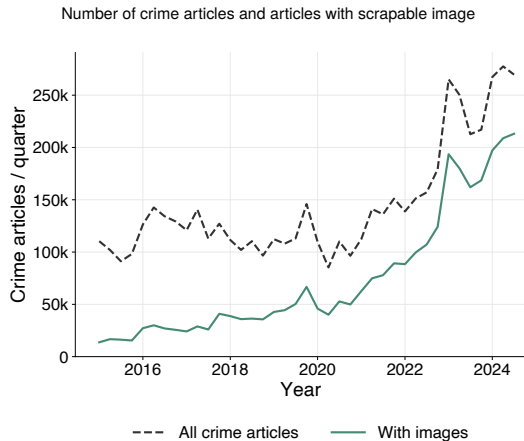
3. Initial findings

Dataset overview

Outlets	Count
Distinct outlets	1,875
Type	
website	691
newspaper	571
tv	334
radio	145
magazine	124
wire service	10
Classified local or national	1,875
local	1,077
national	798
With any mugshot	1,594

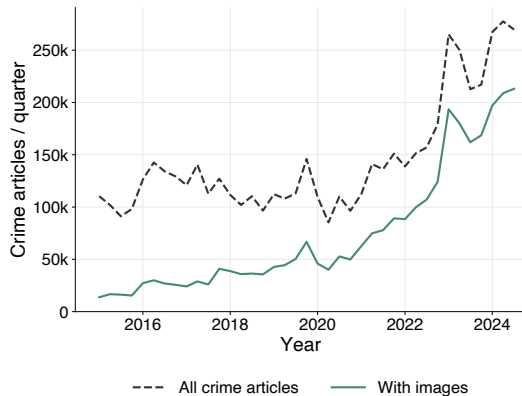
Articles	Count
Crime articles	6,217,186
with images	3,026,718
with mugshot	295,257
Articles per outlet	
Mean	3,304
Median	729

Article, image and mugshot prevalence

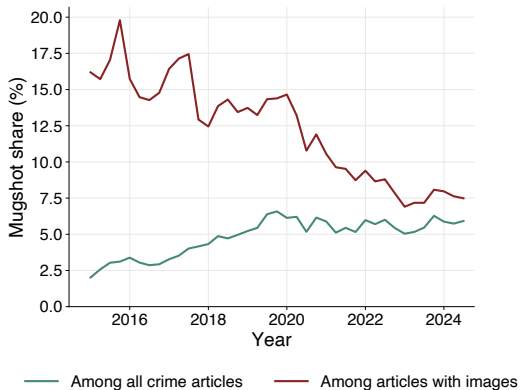


Article, image and mugshot prevalence

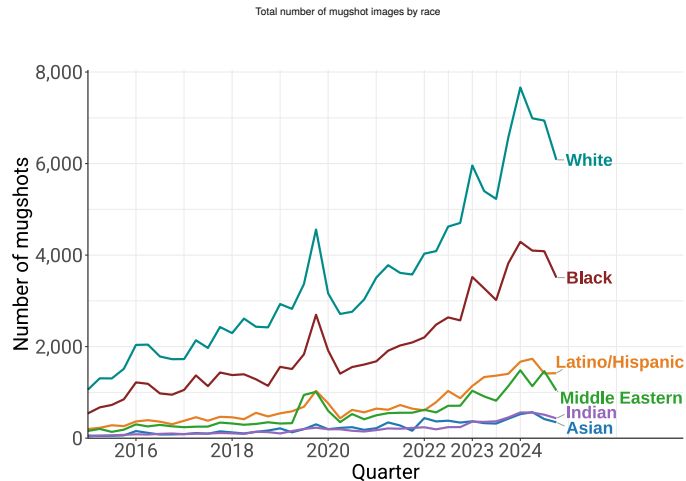
Number of crime articles and articles with scrapable image



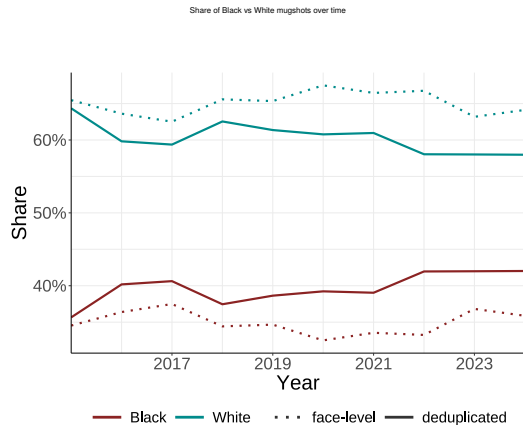
Share of articles with mugshot image over time



Black and White individuals account for the majority of mugshots



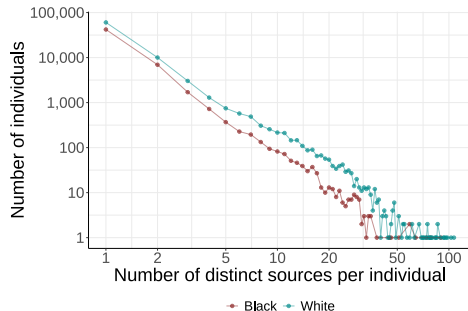
Deduplication increases share of Black individuals in mugshots



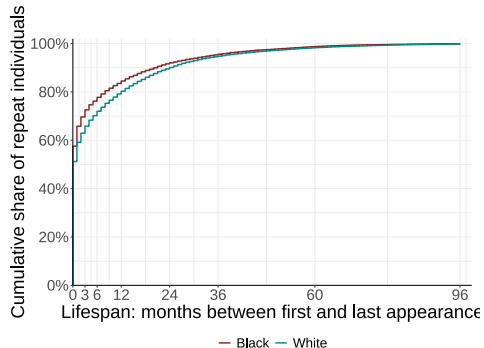
Dotted: total mugshot images; Solid: unique individuals in mugshots

Black individuals get covered less frequently and within a shorter time-span

Number of distinct sources per individual's mugshot by Black vs White (log-log)



Spread before individual's first and last appearance in a news mugshot



Within source and geographical spread

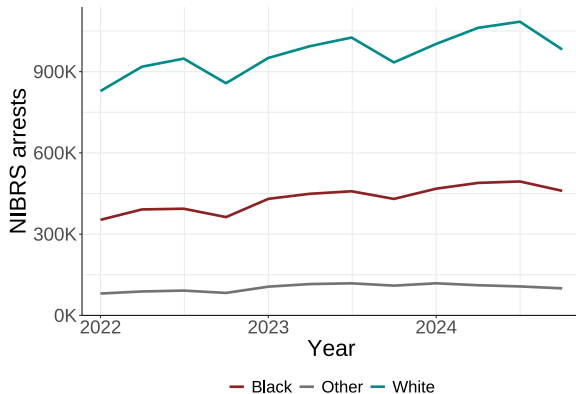
Initial Findings

- Most mugshots depict White persons, followed by Black individuals.
- Black mugshots tend more often to be singletons, are shown in fewer sources, and appear over a shorter time span.

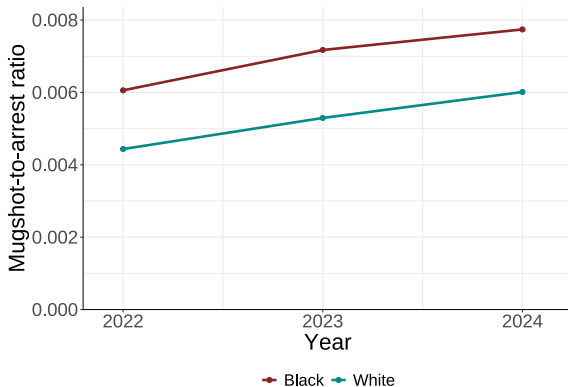
Initial Findings

- Most mugshots depict White persons, followed by Black individuals.
- Black mugshots tend more often to be singletons, are shown in fewer sources, and appear over a shorter time span.
- But are Black individuals over-/under-represented?
- To answer this, we draw on arrest data from the National Incident-Based Reporting System published by the FBI.
- The data contain individual-level information on incidents and arrestees, including race, for police stations covering more than 74% of the U.S. population between 2022 and 2024.

FBI NIBRS arrests overview

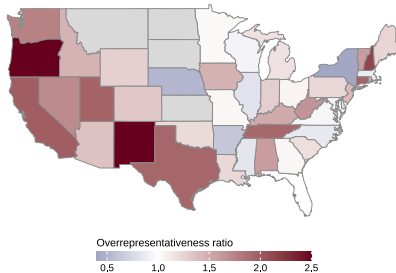


Black mugshots are overrepresented



Black mugshots are overrepresented mostly in the western states

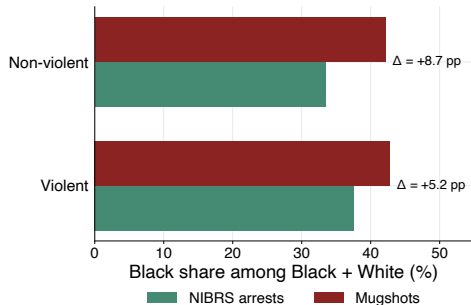
Representation of Black vs. White individuals in mugshots vs. arrests by region



Red: overrepresentation of Black mugshots relative to arrests; blue: overrepresentation of White mugshots relative to arrests. Grey: insufficient data. The representation is shown as $(black_mugshots/black_arrests)/(white_mugshots/white_arrests)$. Data is filtered for local news and years 2022–2024.

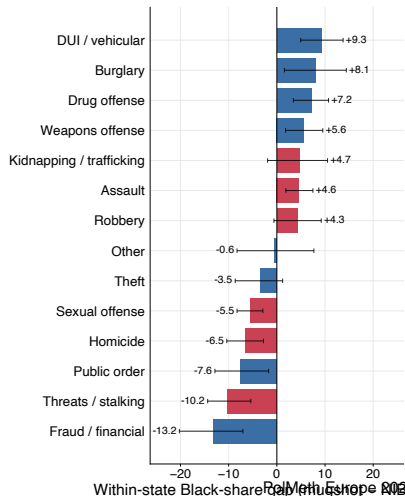
Black individuals are overrepresented especially for non-violent crimes.

Share of Black individuals in news mugshots vs arrests: violent vs non-violent crimes



Red: Black share in mugshot articles, green: Black share in NIBRS arrests. Data restricted to 2022–2024.

Gap between black-share in news mugshots versus arrests within states by crime type



Does Limiting Mugshots Publications Increase or Reduce Disparities? (Preliminary)

- Some States introduced bans on mugshot publication

State	Date of Ban
New York	March 2019
Utah	July 2021
Oregon	January 2022
Lousiana	August 2022

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(Preliminary)

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$$\frac{Mugshot_{st}^G}{Arrest_{st}^G} = \beta^G MugshotPolicy_s \times Post_t + \alpha_s + \alpha_t + \varepsilon_{st}$$

- $\frac{Mugshot_{st}^G}{Arrest_{st}^G}$: mugshot-to-arrest ratio for group $G = \{\text{Black, White}\}$
- α_s, α_t : state and month fixed effects
- ε_{st} clustered at the state level
- β^G : effect of mugshot policies on the mugshot-to-arrest ratio

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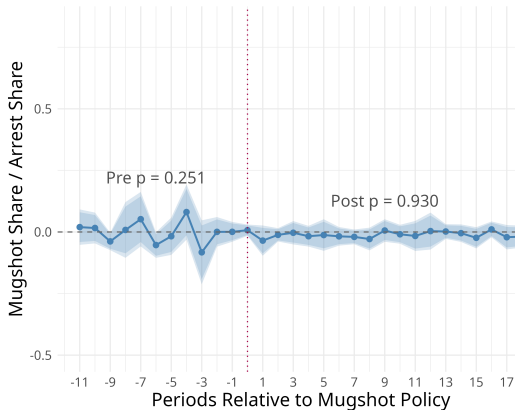
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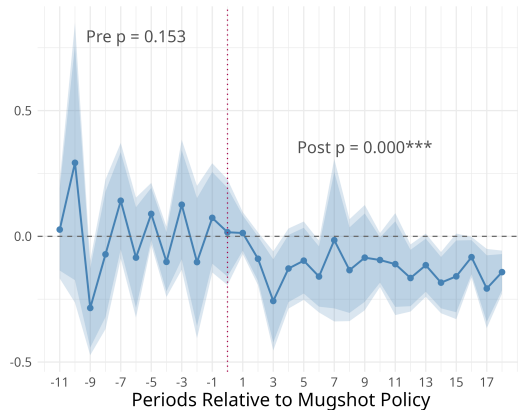
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Policies limiting mugshots publications decrease racial disparities (Preliminary)

White: $\beta^W \simeq 0$



Black: $\beta^B < 0$



Effect of Mugshot Policies Tables

	Log(Mugshot-to-Arrest Ratio)					
	Utah		Louisiana		Oregon	
	Black	White	Black	White	Black	White
Mugshot Ban $\times$ Post	-0.181*** (0.013)	-0.043*** (0.010)	-0.054*** (0.009)	-0.020** (0.008)	-0.004 (0.011)	-0.069*** (0.008)
$\beta_B - \beta_W$	-0.138*** (0.016)		-0.035*** (0.012)		0.065*** (0.014)	
Month FE	X	X	X	X	X	X
Observations	3280	3284	2717	2721	3280	3284
R ²	0.238	0.340	0.246	0.330	0.230	0.341
Mean Dep. Var.	0.146	0.112	0.146	0.112	0.146	0.112
State FE	X	X	X	X	X	X

Conclusion and next steps

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- We collect a large data set of crime images from U.S. news articles.
- We investigate the prevalence and distribution of mugshots.
- Black individuals are relatively overrepresented compared to arrests.
- Black individuals are relatively more overrepresented in non-violent crime cases.

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Next steps

- Analyze other features of mugshot images in newspapers
- Investigate what determines the editorial decision to publish a mugshot and its relation to the article text

References

-  Abrams, David S., Marianne Bertrand, and Sendhil Mullainathan (2012). “Do Judges Vary in Their Treatment of Race?” In: *The Journal of Legal Studies* 41.2, pp. 347–383. doi: 10.1086/666006  URL: <https://www.jstor.org/stable/10.1086/666006> (visited on 01/24/2024).
-  Anwar, Shamena, Patrick Bayer, and Randi Hjalmarsson (May 2012). “The Impact of Jury Race in Criminal Trials*”. In: *The Quarterly Journal of Economics* 127.2, pp. 1017–1055. doi: 10.1093/qje/qjs014  URL: <https://doi.org/10.1093/qje/qjs014> (visited on 03/21/2024).
-  Arnold, David, Will Dobbie, and Crystal S Yang (Nov. 2018). “Racial Bias in Bail Decisions*”. en. In: *The Quarterly Journal of Economics* 133.4, pp. 1885–1932. doi: 10.1093/qje/qjy012  URL: <https://academic.oup.com/qje/article/133/4/1885/5025665> (visited on 12/30/2022).
-  Ash, Elliott, Ruben Durante, Maria Grebenshchikova, and Carlo Schwarz (Oct. 2021). *Visual Representation and Stereotypes in News Media*. en. SSRN Scholarly Paper. Rochester, NY. doi: 10.2139/ssrn.3934858  URL: <https://papers.ssrn.com/abstract/3934858> (visited on 02/27/2025).

Appendix

Media slant and its effect

1. News content is **systematically slanted** by ideological, political, and commercial incentives.
2. Slant has **measurable effects** on public perception, attitudes, and downstream behaviour.
3. Its impact is **disproportionately borne by marginalized groups** — whose treatment in coverage shapes how the broader public views them.

Media narratives and marginalized groups (1/2)

- Narratives matter especially for marginalized groups.
- News about immigrant crime increases anti-immigrant / anti-Muslim voting behaviour (Couttenier et al., 2021).
- Slanted terminology (e.g., “illegal immigrant”) shifts political beliefs on immigration (Djourelouva, 2023).

More

Media narratives and marginalized groups (2/2)

- Realistic reflection of suspects' origin in news can reduce anti-immigrant attitudes (Keita, Renault and Valette, 2021).
- Media slant affects sentencing both temporarily (Philippe and Ouss, 2018) and persistently (Ash and Poyker, 2023).
- The identity of the reporter on domestic violence shifts help-seeking behaviour (Marangon, 2024).
- Images of Black people are underrepresented in news overall but overrepresented in crime coverage (Ash et al., 2022).

Mugshot detection — details

1. A vision–language model (GPT-4o-mini) labels a sample of crime-article images as containing any mugshot vs. not; sample manually validated.
2. A face detection model (FaceNet) extracts face crops from this sample of images.
3. An EfficientNet-B0 model is fine-tuned on those labels and applied to the full image corpus.

Face-level model (EfficientNet-B0, $n_{\text{test}} = 300$, $\tau = 0.5$):

accuracy 0.947 | precision 0.927 | recall 0.966 | F1 0.946

Race classification — details

- Faces detected with MTCNN (FaceNet).
- Per-face race classifier, trained on FairFace-style categories: White, Black, Asian, Indian, Latino-Hispanic, Middle Eastern.
- Race per article = majority vote across mugshot faces in the cluster.

	Precision	Recall
Gender (Aggregate)	0.912	
Male	0.914	0.918
Female	0.910	0.906
Ethnicity (Aggregate)	0.731	
White	0.701	0.746
Black	0.852	0.844
Hispanic	0.490	0.493
Asian	0.891	0.909
Indian	0.701	0.669
Middle Eastern	0.576	0.521

Face deduplication — details

- Faces embedded with InceptionResnetV1 (FaceNet, VGGFace2 weights, 512-d).
- Chinese-Whispers clustering with cosine threshold $t = 0.85$, $k = 20$ neighbours.

Appearances per individual	Count
1 (singleton)	135,209 (72%)
2–5	45,243 (24%)
6–20	6,818
21–100	1,551
>100	15

Stage classifier — details

- LLM (GPT-5-mini) prompted to label a 10K-article sample with one of 8 stage classes: *investigation, arrest/charging, pretrial, trial/verdict, sentencing, appeal, corrections/release, aftermath*.
- Sample checked manually.
- Distilled to TF-IDF + multinomial logistic regression and applied to the whole corpus (tested against XGB, RandomForest, and word count inputs).
Stage classifier (8-class, TF-IDF + LogReg, $n_{\text{test}} = 776$):
accuracy 0.746 | macro-F1 0.682

Crime-type classifier — details

- LLM-label-then-distill recipe, multi-label setting with 20 classes (homicide, assault, sexual offense, robbery, burglary, theft, drug, weapons, fraud, DUI, kidnapping/trafficking, public order, cyber crime, threats/stalking, obstruction/perjury, ...).
- Sample checked manually.
- One-vs-rest logistic regression on TF-IDF features (tested against XGB, RandomForest, and word count inputs).

Crime type classifier (20-class multi-label, OvR LogReg + TF-IDF, $n_{\text{test}} = 776$):

micro-F1 0.742 | macro-F1 0.651 | mean ROC-AUC 0.949

primary-type top-1 accuracy 0.754; top-3 0.948 | homicide F1 = 0.92

- Headline classifier figure on this deck uses the **top-1** prediction; a multi-label sensitivity is straightforward.

Named entity recognition — details

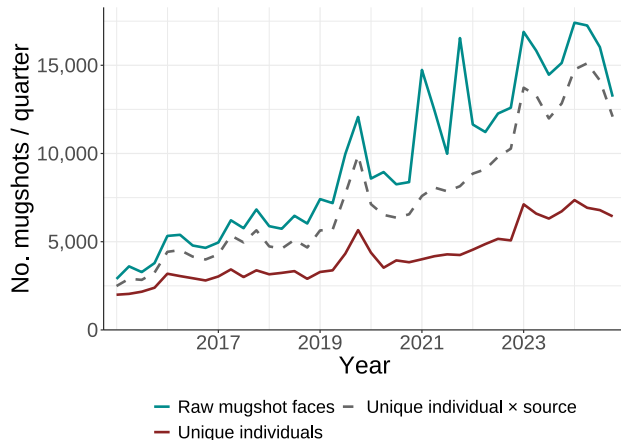
- Per-article NER over title + body for PERSON / LOCATION mentions.
- Used for (a) de-identifying suspects/victims in cluster summaries, and (b) cross-validating face dedup against name dedup on the most-republished individuals.

Outlet geography — details

- GPT classifies each outlet as news, radio, tv, blog, other and local vs. national; outlet HQ city/state/county extracted via GPT with a web-search tool.
- Provides the geographic match needed to compare each outlet's mugshot composition to NIBRS arrests in the same state and county.

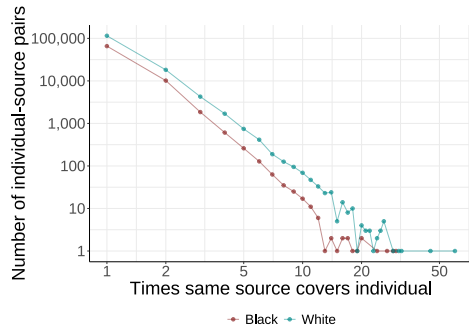
Individuals occur frequently

Total number of mugshot images vs. unique individuals vs. within source

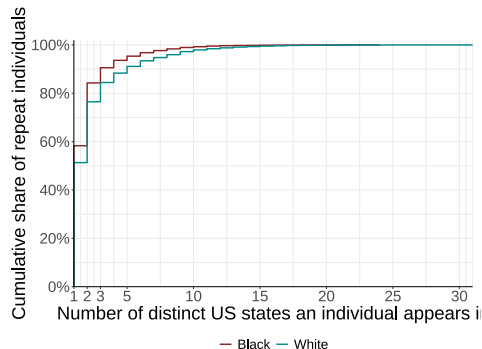


Within source coverage of individuals and geographical spread

Number of an individual's mugshot occurrences for repeated coverage within a source (log-log)



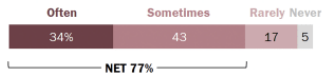
Distribution of the number of distinct states in an individual's mugshot coverage



Crime coverage is a core news topic

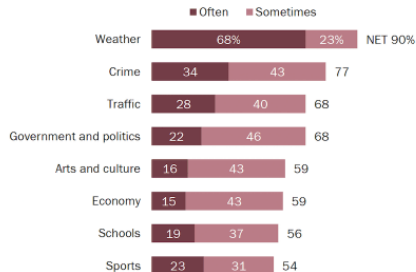
77% of Americans get news about local crime at least sometimes ...

% of U.S. adults who say they ___ get local crime news



Many Americans follow news about local weather, crime, traffic and government

% of U.S. adults who get news and information about each local topic



Source: Pew Research Center survey of U.S. adults conducted Jan. 22-28, 2024.
"Americans' Changing Relationship With Local News"

PEW-KNIGHT INITIATIVE